

HW

Q-1 Data = 0000 0000 0000 0000 0000 0000 0000 0000
1100 0000 0011 0000 0011 0000 0001 0100

- How many bits can you store in 1 memory location/cell?
- Show how the given Data will be stored in memory of a risc-V machine. Assume the starting memory location is 0
- Show how the given Data will be stored in memory of a machine that follows big endian. Assume the starting memory location is 8 approach

Q-2 Data = (AB13529CD)₁₆

- How many bits can you store in 1 memory location/cell?
- Show how the given Data will be stored in memory of a risc-V machine. Assume the starting memory location is 0.
- Show how the given Data will be stored in memory of a machine that follows big endian. Assume the starting memory location is 12. approach

Q-3 Array A is a Byte array with size of 50. Base address of A is stored in X23. Variable g & h are associated with X5, X6 registers. Write the necessary risc-V code of the following code.

$$A[12] = A[5] + A[7] + g + A[2]$$

Q-4 Data Memory: (little endian)

Location	Data	Loc.	Data	Loc.	Data	Loc.	Data	$X_{21} = (12)_{10}$
21	$(AC)_{16}$	24	$(D0)_{16}$	27	$(00)_{16}$	30	$(00)_{16}$	$X_{22} = (13)_{10}$
22	$(12)_{16}$	25	$(FE)_{16}$	28	$(00)_{16}$	31	$(F9)_{16}$	
23	$(05)_{16}$	26	$(23)_{16}$	29	$(00)_{16}$	32	$(23)_{16}$	

Instruction:

LD $X_5, 10(X_{22})$

After executing the above instruction, what is the value of X_5 ?

Q-5 Data Memory: (little endian)

Location	Data	Loc.	Data	Loc.	Data	Loc.	Data	$X_{21} = (12)_{10}$
21	0011 1011	24	0000 1111	27	1111 1111	30	0011 0000	$X_{22} = (13)_{10}$
22	0011 0000	25	1011 0000	28	1011 0011	31	1000 0001	
23	0001 1111	26	0000 1011	29	0001 1000	32	1000 1111	

Instruction:

SB $X_5, 10(X_{22})$

$X_5 =$
 $\overset{\text{msb}}{0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000}$
 $0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 1011\ 1010$
 $\underset{\text{lsb}}{}$

After executing the above instruction, update the data memory.